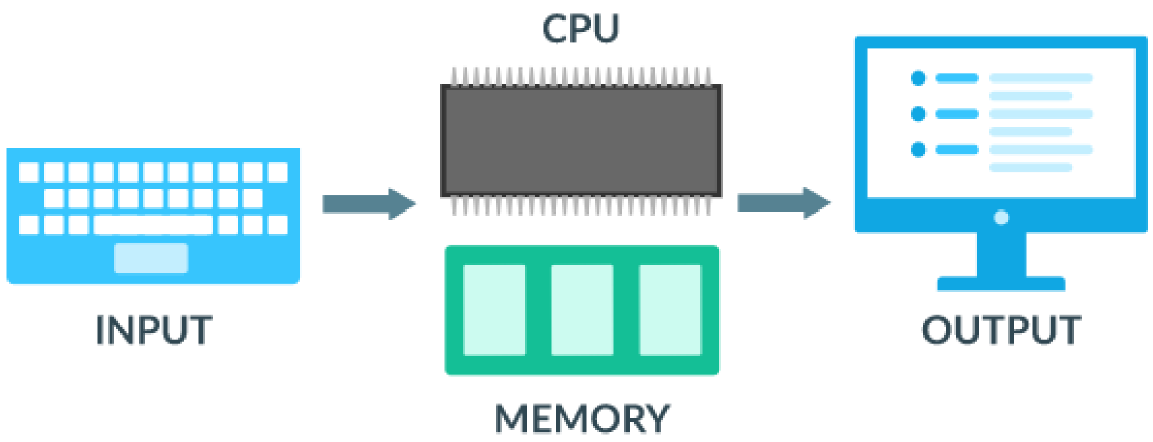
# Computer Components

An Overview

Adapted from Khan Academy’s Unit on Computers

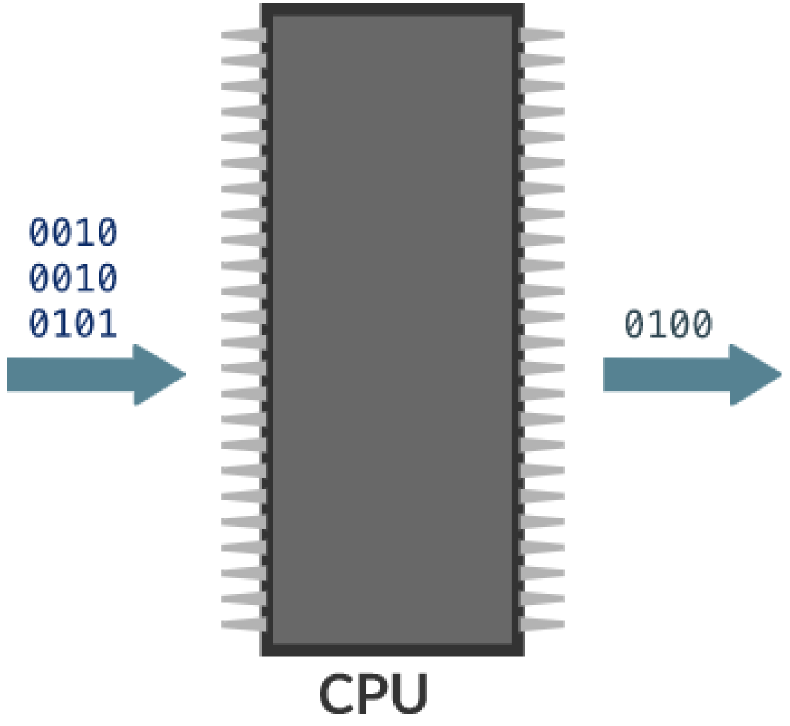
# What are the parts of a computer?

- At a high level, all computers are made up of a processor (CPU), memory, and input/output devices.
- Each computer receives **input** from a variety of devices, processes that data with the **CPU** and **memory**, and sends results to some form of **output**.



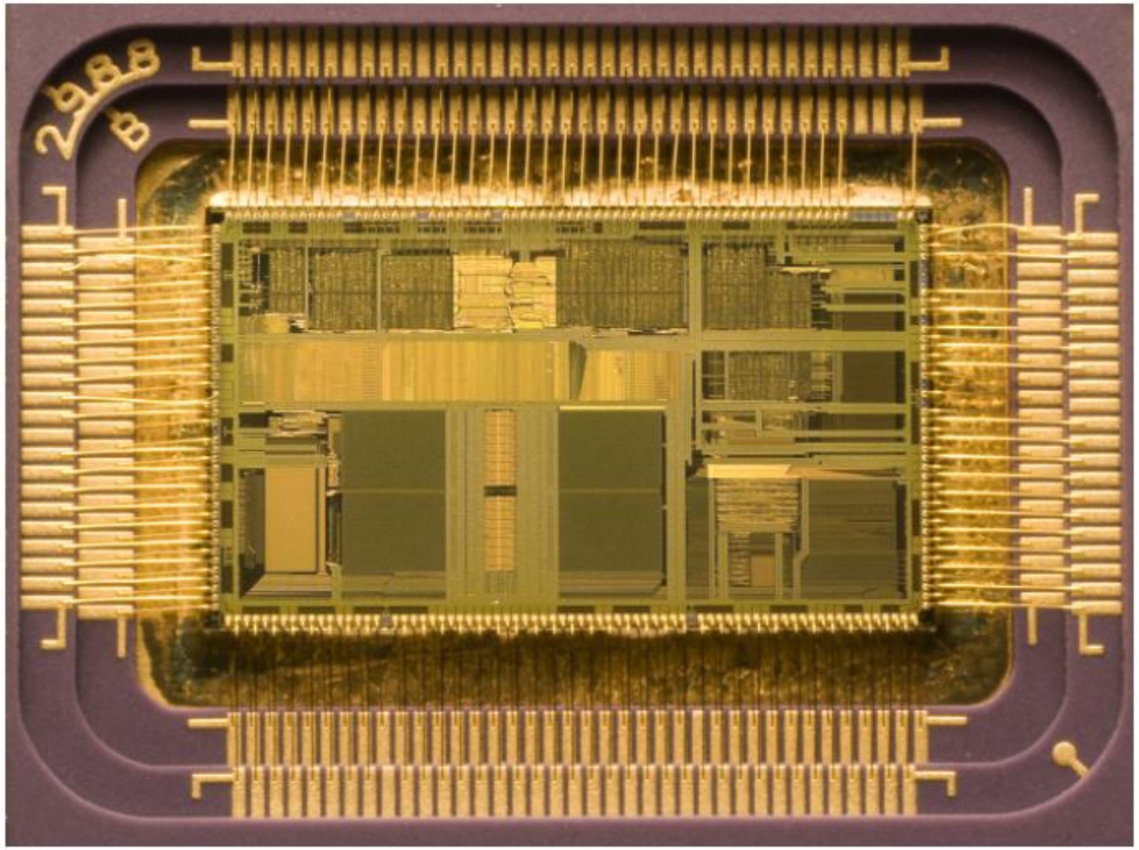
# Central Processing Unit (CPU)

- The **CPU** is the brain of a computer, containing all the circuitry needed to process input, store data, and output results.
- The CPU is constantly following instructions of computer programs that tell it which data to process and how to process it. Without a CPU, we could not run programs on a computer.



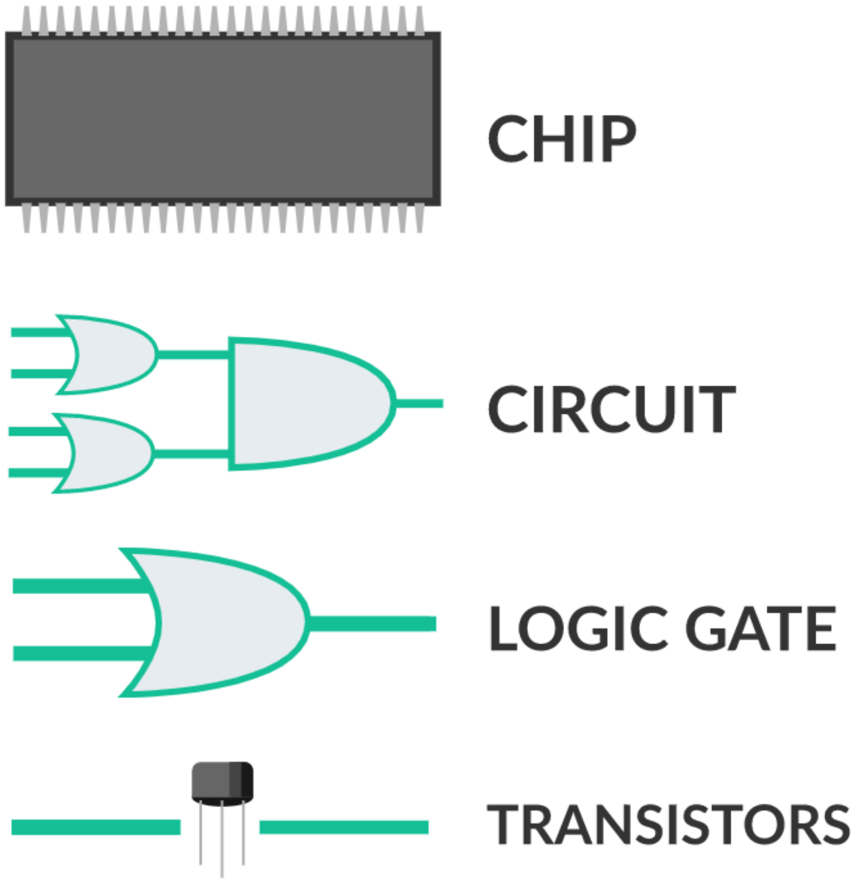
# Inside the CPU

- At the hardware level, a CPU is an **integrated circuit**, also known as a **chip**. An integrated circuit "integrates" millions or billions of tiny electrical parts, arranging them into circuits and fitting them all into a compact box.



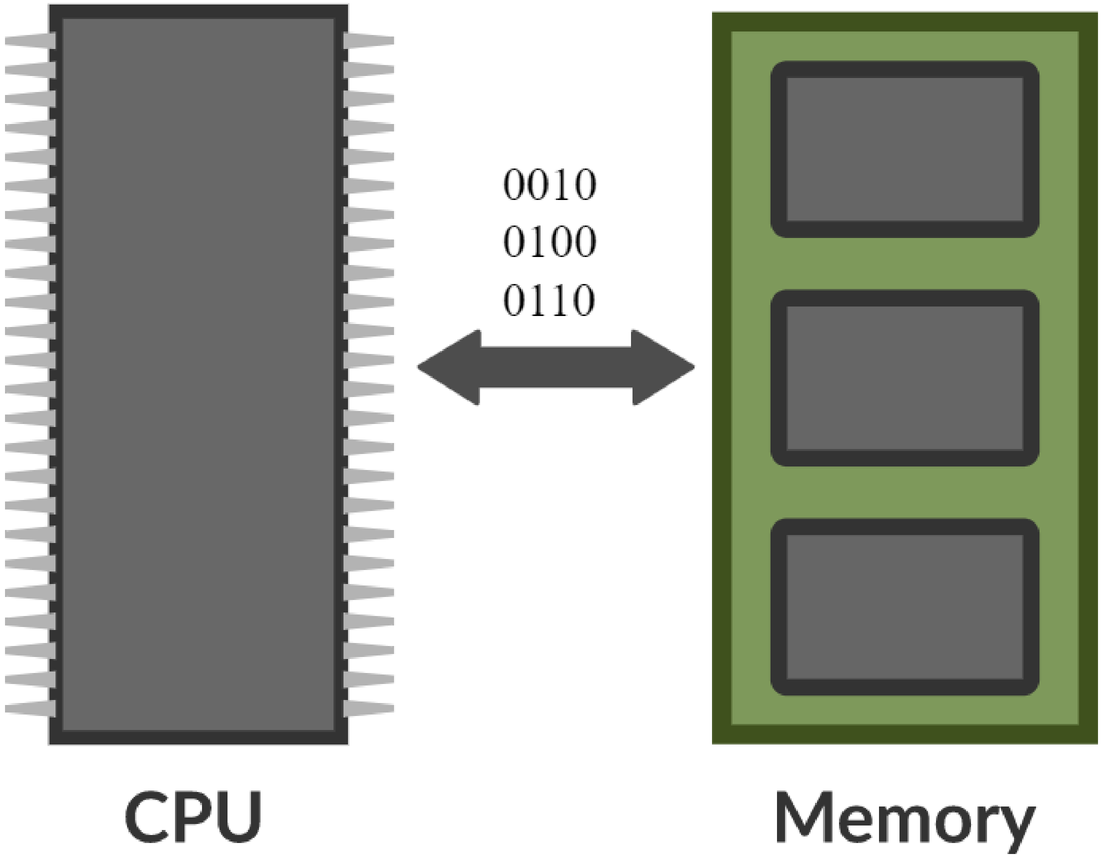
# Inside the CPU

- We can visualize the layers of the CPU chip:
- Some of those layers are **physical devices**, like the chip and transistors, and some of those layers are **abstractions**, like logic circuits and gates.
- It's impressive that we can put together seemingly simple devices like logic gates to create CPUs that power complex devices like our phones, computers, and even self-driving cars.



# Computer memory

- When input devices send binary data to a CPU, it immediately stores that data in **memory** to make it easier to process.
- Since the CPU is constantly using data from memory, they're connected via a **memory bus**, a high speed communication transfer system, typically made from wires, conductors, or optical fibers.



# Secondary memory

- When computers need to store data for long-term retrieval, they use **secondary memory**, also known as auxiliary storage or external memory.
- Examples:
  - Hard disk drive (HDD)
  - Solid State drive (SSD)
  - USB drive

# From electricity to bits

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- In a computer, information travels over wires. The easiest way to convey information in a wire is to consider it "on" or "off", based on how much electricity is going through it.


- An "on" wire represents 1, and an "off" wire represents 0.


- This small piece of information is called a **"bit"**, and it's the smallest piece of information that computers process.

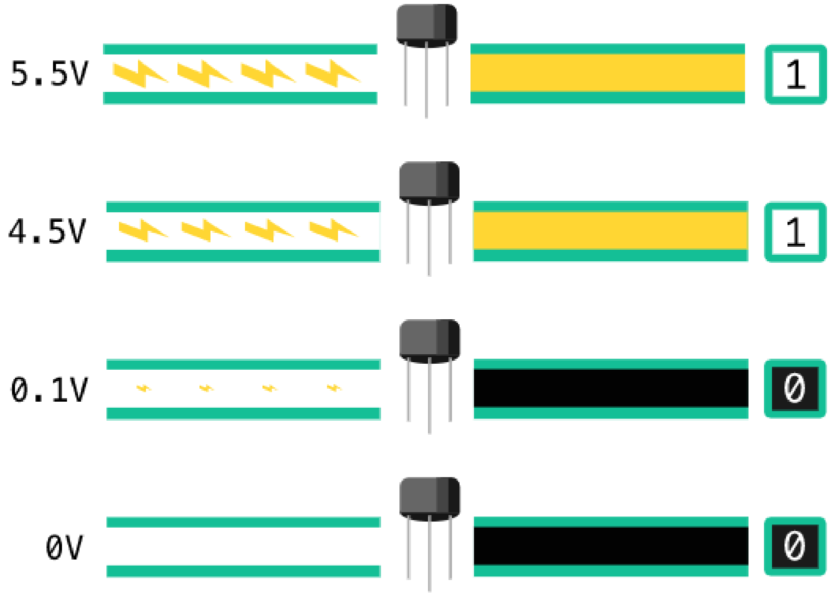
# Behind the abstraction

- In actuality, a wire isn't exactly "on" or exactly "off". That's an **abstraction** that simplifies the details of how computers work.

- A wire can have varying amounts of electricity flowing through it, but a computer needs to be able to interpret the electricity in a wire as *either* **definitely** 0 or **definitely** 1.

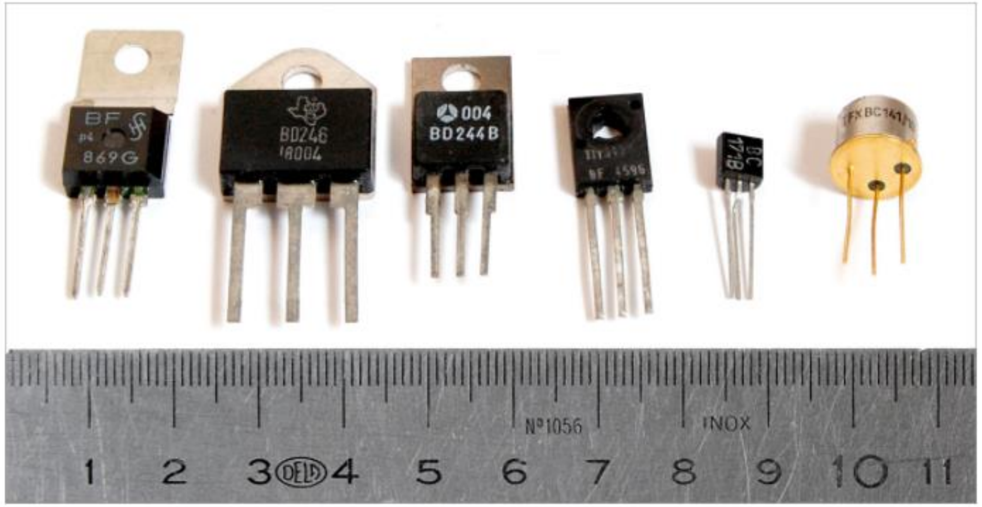
# Behind the abstraction

- How much electricity is "enough"? That depends on the transistor and its threshold voltage. If an engineer uses a transistor with a threshold voltage of 4.5 volts, then any voltage of 4.5 or higher will turn the transistor on. At lower voltages, the transistor stays off.

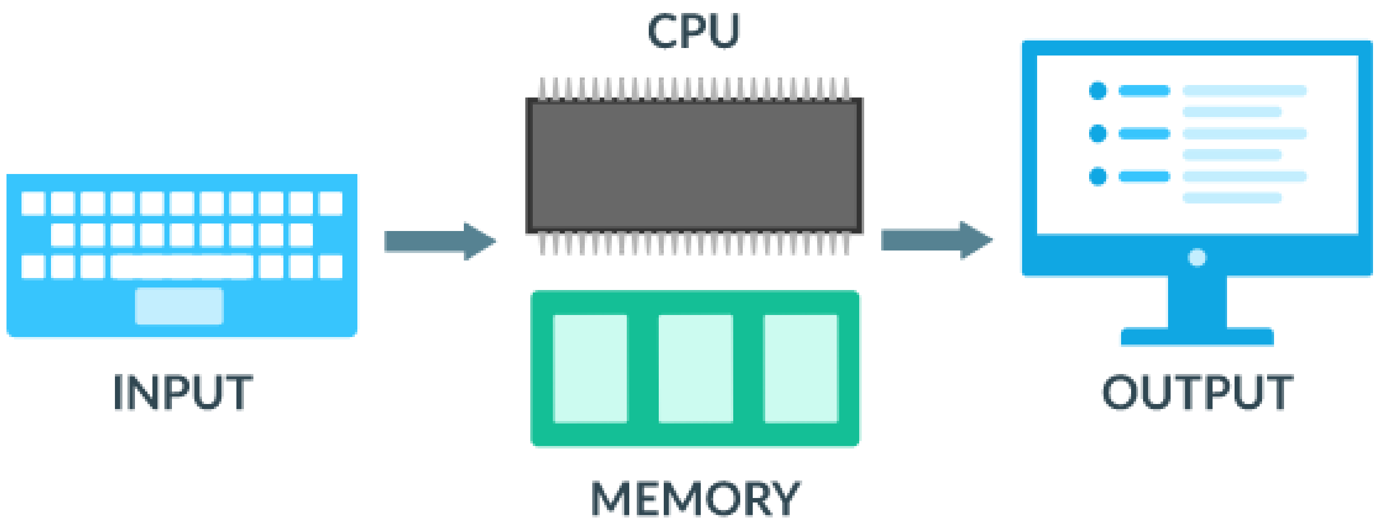


# Behind the abstraction

- There's a huge variety in transistors. Engineers choose the transistors that are the best fit for the job, by considering characteristics like the threshold voltage, material, and size.
- The transistors inside your computer are so small, we would need a high-powered microscope to see them. However, transistors are also used in other electrical projects and those transistors are ones you could pick up with your fingers. A few examples are to the right:



Physical devices:




Abstraction: